

Student Performance Analysis System

1. Aim : To design and implement a Student Performance Analysis System in Java using Object-Oriented Programming concepts to calculate total marks, average, percentage, grade, highest and lowest marks, pass/fail status, and overall academic performance.

2. Description : The Student Performance Analysis System is a Java application designed for colleges to analyze a student's academic performance.

The system accepts the student's roll number, name, and marks in five subjects. It uses an array to store marks and performs various calculations such as total marks, average marks, highest marks, lowest marks, and percentage. The system assigns a grade according to the percentage and determines whether the student has passed or failed. String operations such as trim(), toUpperCase(), and length() are used to format and process the student's name. The Math.round() method is used to round the percentage to two decimal places. The program also displays a performance remark based on the student's grade.

Steps :

1. Start the program.
2. Import the Scanner class.
3. Create a Student class.
4. Declare the data members:
 - rollNo
 - name
 - marks[]
5. Create a constructor to initialize the student details.
6. Create the calculateTotal() method to calculate the total marks using the array.
7. Create the calculateAverage() method to calculate the average marks.
8. Create the findHighest() method using Math.max() to find the highest mark.
9. Create the findLowest() method using Math.min() to find the lowest mark.
10. Create the calculatePercentage() method to calculate the percentage.
11. Create the getGrade() method to assign a grade according to the percentage.
12. Create the getRemark() method to display the performance remark.

13. Create the displayDetails() method.
14. Use trim(), toUpperCase(), and length() to process the student's name.
15. Use Math.round() to round the percentage to two decimal places.
16. Create a Student object in the main() method.
17. Accept the roll number, name, and marks of five subjects from the user.
18. Store the five subject marks in an integer array.
19. Call displayDetails() to display the complete performance report.
20. Close the Scanner.
21. Stop the program.

Algorithm :

- 1. Start the program.**
- 2. Import the Scanner class.**
- 3. Create a class named Student.**
- 4. Declare the data members:**
 - **Roll number**
 - **Student name**
 - **Array to store marks of 5 subjects**
- 5. Create a constructor to initialize the roll number, name, and marks.**
- 6. Create the calculateTotal() method:**
 - **Initialize total = 0.**
 - **Add all subject marks using a loop.**
 - **Return the total marks.**
- 7. Create the calculateAverage() method:**
 - **Divide total marks by the number of subjects.**
 - **Return the average marks.**
- 8. Create the findHighest() method:**
 - **Assume the first mark is the highest.**
 - **Compare all marks using Math.max().**

- Return the highest mark.

9. Create the findLowest() method:

- Assume the first mark is the lowest.
- Compare all marks using Math.min().
- Return the lowest mark.

10. Create the calculatePercentage() method:

- Calculate percentage using:

Percentage = (Total Marks / Maximum Marks) × 100

- Return the percentage.

11. Create the getGrade() method:

- If percentage ≥ 90, assign A+.
- If percentage ≥ 80, assign A.
- If percentage ≥ 70, assign B.
- If percentage ≥ 60, assign C.
- If percentage ≥ 50, assign D.
- Otherwise, assign F.

12. Create the getRemark() method:

- Display a performance remark according to the grade.

13. Create the displayDetails() method:

- Remove extra spaces from the name using trim().
- Convert the name to uppercase using toUpperCase().
- Find the name length using length().
- Round the percentage to two decimal places using Math.round().
- Display all student performance details.

14. In the main() method, create a Scanner object.

15. Read the student's roll number and name.

16. Create an integer array of size 5.

17. Read and store marks of five subjects in the array.

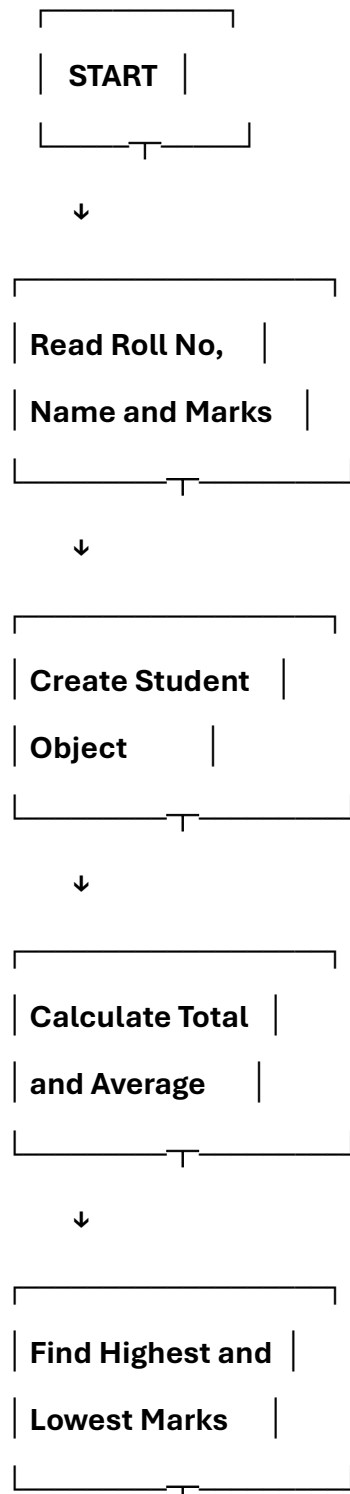
18. Create a Student object using the entered details.

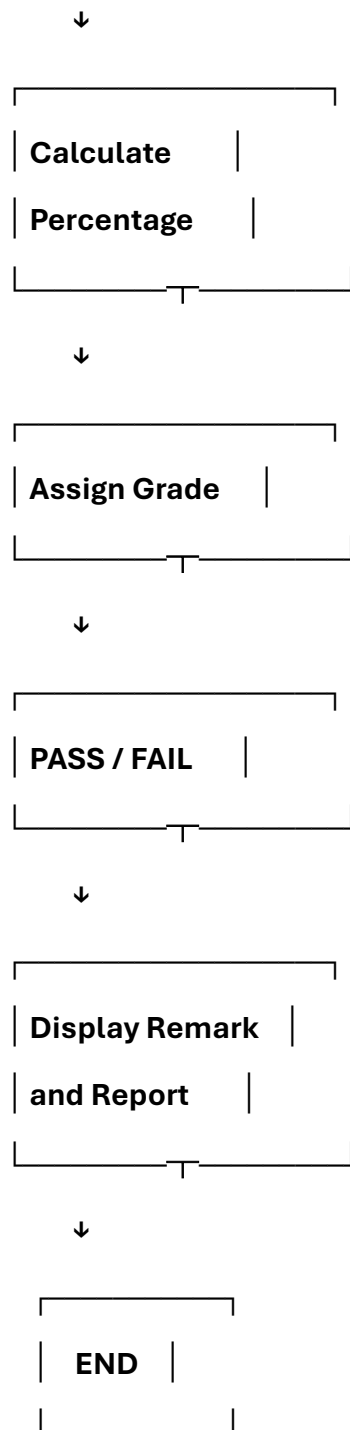
19. Call displayDetails() to display the complete performance report.

20. Close the Scanner.

21. Stop the program.

UML DIAGRAM :





CODE :

```
import java.util.Scanner;

class Student {

    private int rollNo;

    private String name;
```

```
private int[] marks;
```

```
Student(int rollNo, String name, int[] marks) {
```

```
    this.rollNo = rollNo;
```

```
    this.name = name;
```

```
    this.marks = marks;
```

```
}
```

```
int calculateTotal() {
```

```
    int total = 0;
```

```
    // Calculate total marks using the array
```

```
    for (int mark : marks) {
```

```
        total += mark;
```

```
    }
```

```
    return total;
```

```
}
```

```
double calculateAverage() {
```

```
    return (double) calculateTotal() / marks.length;
```

```
}
```

```
int findHighest() {
```

```
    int highest = marks[0];
```

```
    // Find the highest mark
```

```
    for (int mark : marks) {
```

```
        highest = Math.max(highest, mark);
```

```
    }
```

```
    return highest;
}
```

```
int findLowest() {
    int lowest = marks[0];

    // Find the lowest mark
    for (int mark : marks) {
        lowest = Math.min(lowest, mark);
    }

    return lowest;
}
```

```
double calculatePercentage() {
    return ((double) calculateTotal() / (marks.length * 100)) * 100;
}
```

```
String getGrade() {
    double percentage = calculatePercentage();

    if (percentage >= 90)
        return "A+";
    else if (percentage >= 80)
        return "A";
    else if (percentage >= 70)
        return "B";
}
```

```
    else if (percentage >= 60)
        return "C";
    else if (percentage >= 50)
        return "D";
    else
        return "F";
}
```

```
String getRemark() {
    String grade = getGrade();

    switch (grade) {
        case "A+":
            return "Outstanding Performance";
        case "A":
            return "Excellent Performance";
        case "B":
            return "Very Good Performance";
        case "C":
            return "Good Performance";
        case "D":
            return "Satisfactory Performance";
        default:
            return "Needs Improvement";
    }
}
```

```
void displayDetails() {
```

```

String formattedName = name.trim().toUpperCase();

// Round percentage to two decimal places
double percentage = Math.round(calculatePercentage() * 100.0) / 100.0;

System.out.println("\n----- STUDENT PERFORMANCE REPORT -----");
System.out.println("Roll Number   : " + rollNo);
System.out.println("Student Name   : " + formattedName);
System.out.println("Name Length    : " + formattedName.length());
System.out.println("Total Marks    : " + calculateTotal());
System.out.println("Average Marks   : " + calculateAverage());
System.out.println("Highest Mark    : " + findHighest());
System.out.println("Lowest Mark     : " + findLowest());
System.out.println("Percentage      : " + percentage + "%");
System.out.println("Grade          : " + getGrade());

if (percentage >= 50)
    System.out.println("Result        : PASS");
else
    System.out.println("Result        : FAIL");

System.out.println("Remark        : " + getRemark());
}
}

public class Studentperformance{

    public static void main(String[] args) {

```

```
Scanner sc = new Scanner(System.in);

System.out.print("Enter Roll Number: ");
int rollNo = sc.nextInt();
sc.nextLine();

System.out.print("Enter Student Name: ");
String name = sc.nextLine();

int[] marks = new int[5];

System.out.println("Enter marks for 5 subjects:");

for (int i = 0; i < marks.length; i++) {
    System.out.print("Subject " + (i + 1) + ": ");
    marks[i] = sc.nextInt();
}

Student student = new Student(rollNo, name, marks);

student.displayDetails();

sc.close();
}
}
```

OUTPUT :

```
32      }  
...  
Problems 7  Output  Debug Console  Terminal  Ports  
ges' '-cp' 'C:\Users\yabhi\AppData\Roaming\Code\User\workspaceStorage\368729c680f0f32712f9803f080a0563\redhat.j  
559b\bin' 'Studentperformance'  
Enter Roll Number: 264  
Enter Student Name: Abhinay  
Enter marks for 5 subjects:  
Subject 1: 95  
Subject 2: 78  
Subject 3: 80  
Subject 4: 88  
Subject 5: 90  
  
----- STUDENT PERFORMANCE REPORT -----  
Roll Number      : 264  
Student Name     : ABHINAY  
Name Length      : 7  
Total Marks      : 431  
Average Marks    : 86.2  
Highest Mark     : 95  
Lowest Mark      : 78  
Percentage       : 86.2%  
Grade           : A  
Result          : PASS  
Remark          : Excellent Performance  
PS C:\Users\yabhi\abhi.java>
```